Clinical Guidelines for Epilepsy Management and Status Epilepticus

1. Diagnostic Workup and Neuroimaging

Initial evaluation requires comprehensive clinical history from the patient and eyewitnesses. Mandatory diagnostics include standard 10-20 scalp EEG, sleep-deprived EEG or prolonged video-EEG monitoring to capture paroxysmal events.

Magnetic Resonance Imaging (MRI) with epilepsy protocol (high-resolution T1 volumetric, T2 coronal oblique perpendicular to long axis of hippocampus, FLAIR, and susceptibility-weighted imaging) is standard to detect structural epileptogenic lesions including hippocampal sclerosis, vascular cavernomas, and focal cortical dysplasia.

2. Pharmacological Management: Focal Epilepsy

Monotherapy is the initial strategy. First-line antiseizure medications (ASMs) for focal onset seizures include lamotrigine, levetiracetam, and carbamazepine. Oxcarbazepine and lacosamide represent evidence-based alternatives.

Sodium channel blocking ASMs (carbamazepine, phenytoin, oxcarbazepine) are highly effective in focal seizures but must be avoided in primary generalized epilepsies as they can exacerbate absence and myoclonic seizures.

Lacosamide enhances slow inactivation of voltage-gated sodium channels without impacting fast inactivation, offering favorable cognitive tolerability and intravenous availability.

3. Pharmacological Management: Generalized and Absence Epilepsies

Broad-spectrum ASMs are mandatory for generalized seizures. Sodium valproate provides superior seizure freedom for idiopathic generalized epilepsy and juvenile myoclonic epilepsy, but carries black-box warnings for teratogenicity and cognitive impairment in offspring, severely restricting its use in women of childbearing potential.

Lamotrigine and levetiracetam serve as preferred alternatives in women of childbearing potential. Ethosuximide is the first-line medication of choice for childhood absence epilepsy due to specific T-type calcium channel inhibition in the thalamus, though it has no efficacy against tonic-clonic or focal seizures.

4. Drug-Resistant Epilepsy (DRE) and Surgical Evaluation

Drug-resistant epilepsy is formally defined by the ILAE as failure of adequate trials of two tolerated and appropriately chosen and used ASM schedules (whether as monotherapies or in combination) to achieve sustained seizure freedom.

Patients meeting criteria for drug-resistant epilepsy should be promptly referred to a comprehensive epilepsy center for presurgical evaluation, including video-EEG telemetry, PET/SPECT functional imaging, neuropsychological testing, and potential resective surgery (e.g. anterior temporal lobectomy, lesionectomy) or neuromodulation (Vagus Nerve Stimulation, Responsive Neurostimulation).

5. Status Epilepticus Emergency Protocol

Status epilepticus (SE) is a neurological emergency defined as continuous seizure activity exceeding 5 minutes or two or more distinct seizures without full recovery of consciousness between events.

Phase 1 (Early Status, 0-10 min): Emergency stabilization (ABC), check blood glucose, and administer intravenous benzodiazepines (IV lorazepam 0.1 mg/kg or IM midazolam 10 mg).

Phase 2 (Established Status, 10-30 min): If seizures continue, administer non-sedating

IV second-line ASMs: IV levetiracetam (60 mg/kg, max 4500 mg), IV fosphenytoin (20 mg PE/kg), or IV sodium valproate (40 mg/kg).

Phase 3 (Refractory Status, >30-60 min): Seizures persisting despite second-line therapy require general anesthesia (IV midazolam, propofol, or ketamine) with continuous EEG monitoring in the intensive care unit to titrate toward burst suppression.